

Linear actuator/ Power Screw & Stability

- Stability:

- Max extension:

- ◆ Link's length: 35 cm
- ◆ Max Height: 1.6 m
- ◆ No. of Pairs: 5/Side

$$H = n \cdot L \cdot \sin \theta_m$$

$$\therefore \theta_m = 66^\circ$$

❖ Note: We can add more supports (rods between sides of the scissors) to reach to max of 72 degrees

- Min Collapsing: assuming the max height at collapsing is 30cm

$$H = n \cdot L \cdot \sin \theta_{min}$$

$$\therefore \theta_{min} = 9.8 \cong 10^\circ$$

- Linear actuator:

- Stroke:

want the actuator to move the scissor mechanism by **33 cm** (the maximum height per pair). The actuator moves in the horizontal direction and extends the scissor mechanism diagonally.

- ◆ Link Length (L): 35 cm,
- ◆ Height Change (ΔH): 33 cm.

- At full extension:

$$\sin \theta = \frac{\Delta H}{L} = \frac{33}{35} = 0.943$$

$$\therefore \theta = 70^\circ$$

- The horizontal distance covered by actuator is:

$$d = L \cdot \cos \theta = 35 \times \cos(70) = 11.9 \text{ cm} \cong 12 \text{ cm}$$

2. Placement/ Adjustment:

A. Why **Diagonal Placement** for the Actuator is better:

In a scissor lift, the links rotate around their joints, and the lifting force is applied horizontally by the actuator to extend or retract the links. The diagonal placement of the actuator provides several mechanical advantages:

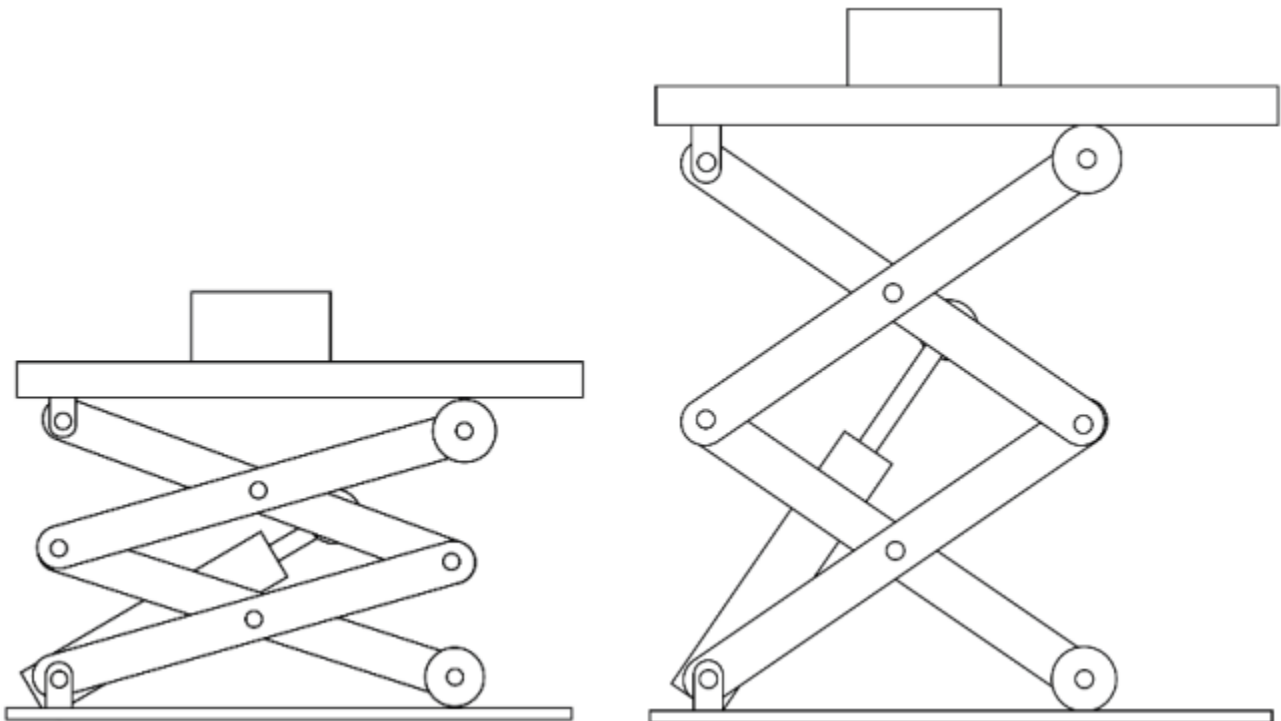
- **Mechanical Leverage:** Placing the actuator diagonally along the links (especially in the first or second pair of links) allows it to use the geometry of the scissor mechanism to reduce the force required to extend the lift. By pushing or pulling the links diagonally, it optimally uses the link length to transfer force more efficiently than a horizontal placement.
- **Torque Reduction:** As the scissor lift extends, the links move from a nearly horizontal to a more vertical position. The diagonal actuator placement helps balance the torque generated as the links rotate. With diagonal placement, less torque is required at the beginning of the lift (when the links are near horizontal) and less force is needed as the links extend upward.

B. Force Analysis:

We will analyze the forces in two different stages of the lift:

1. Starting position (collapsed, links near horizontal),
2. Fully extended position (links near vertical).
 - Link length (L) = 35 cm,
 - Height per pair (ΔH) = 33 cm,
 - Actuator stroke = 12 cm

Let's assume the actuator is mounted **diagonally with an angle α** from the horizontal, which varies as the lift extends. The key force involved here is the actuator force (F_a) needed to lift the load.



- **Collapsed State (Near Horizontal Links):**

In the collapsed state, the scissor links are at a small $\approx 10^\circ$ from the horizontal, the actuator needs to lift the entire system from this position. At this point, the **effective force required is highest** because the scissor mechanism is closest to horizontal, which is the least efficient position for lifting.

Let the load weight (W) on the scissor lift be 60 kg (which is equivalent to 588 N of force due to gravity or ≈ 600 N). The force the actuator needs to apply (F_a) is:

$$F_a = \frac{W}{2 \times \sin \theta_{min}} = \frac{600}{2 \times \sin (10)} = 1727.7 \text{ N}$$

At the collapsed state, the actuator must provide a force of **1730 N** to initiate lifting, which is quite high due to the small angle.

- **Extended State (Near Vertical Links):**

In the extended state, the links form a larger angle $\approx 66.7^\circ$ with the horizontal, making the lifting process much more efficient. Here, the actuator only needs to provide a fraction of the force compared to the collapsed state.

$$F_a = \frac{W}{2 \times \sin \theta_{min}} = \frac{600}{2 \times \sin (66.7)} = 326.64 \text{ N}$$

At the fully extended state, the actuator only needs to provide **327 N** of force, which is significantly lower than the force required in the collapsed state.

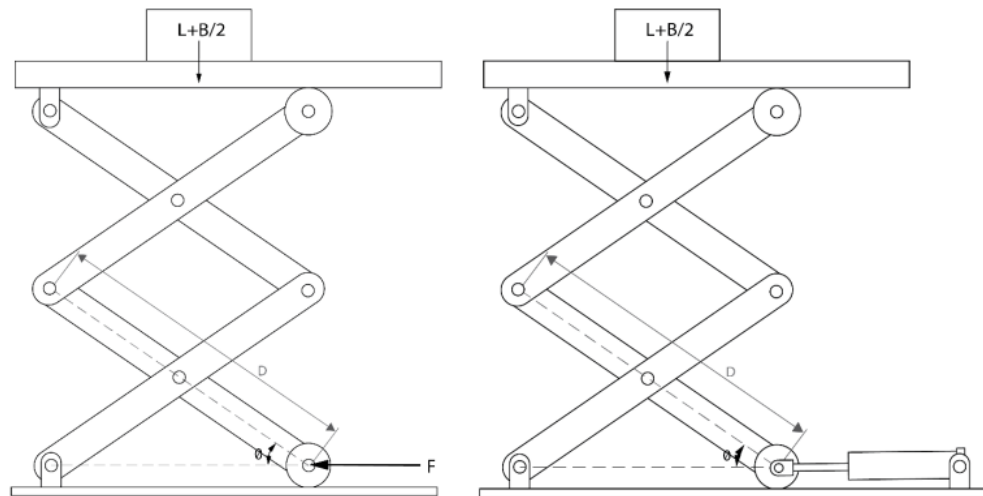
SOOO, the force difference between the collapsed and extended positions highlights the importance of diagonal placement:

- In the collapsed position: The actuator is working at its maximum force because the links are almost horizontal, making the lifting process inefficient.
- In the extended position: The actuator needs much less force to lift because the links are nearly vertical, where the mechanical advantage of the scissor lift is at its highest.

If the actuator were placed purely horizontally, the initial force required would be even higher, as it would be working directly against the weight of the load without the benefit of the scissor mechanism's geometry.

By placing the actuator diagonally, you can reduce the required force at the most critical stages (near the collapsed state) because the diagonal movement makes better use of the scissor mechanism's geometry to lift the load more efficiently.

C. Horizontal Placement and its force analysis:



- Collapsed State (Near Horizontal Links):

$$Fa = \frac{W}{2 \times \sin \theta_{min}} = \frac{600}{2 \times \tan(10)} = 1700.7 \text{ N}$$

- Extended State (Near Vertical Links):

$$Fa = \frac{W}{2 \times \sin \theta_{min}} = \frac{600}{2 \times \tan(66.7)} = 130 \text{ N}$$

SOOO, in the collapsed state, the horizontal actuator needs to provide **1700 N** of force to lift the load, which is similar to the diagonal case but without the mechanical advantage of the diagonal placement.

n the fully extended state, the horizontal actuator requires **130 N** of force to extend the scissor lift, which is lower than the diagonal placement at this stage.

State	Diagonal	Horizontal
Collapsed (10°)	1730 N	1700 N
Extended (70°)	327 N	130 N

- Collapsed State: The force required is nearly the same for both placements (around 1700 N), but diagonal placement has better mechanical leverage, allowing the force to be applied more efficiently. In horizontal placement, the actuator pushes directly against the links horizontally, which increases wear and tear on the system and reduces efficiency.
- Extended State: The force required for the horizontal actuator is slightly lower, but this is only due to the horizontal force being applied directly along the scissor lift's movement direction. However, diagonal placement is still more reliable and efficient in lifting because it allows smoother operation throughout the entire range of motion.